🗠 CAPSTONE PROJECT DOCUMENTATION

Submitted in partial fulfilment of the course requirements

Smart Stock Market Analyst

(Generative AI)

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Subject	Artificial Intelligence & Machine Learning
Academic Year	2025 – 2026
Technology Stack	Python · Streamlit · yFinance · Gemini AI · Plotly

01 Project Name & Overview

Project Title

Smart Stock Market Analyst (Generative AI)
A Python-based web application that combines real-time stock data with Google Gemini AI to deliver intelligent market analysis and news insights.

Technology Used

- Python 3.x — Core programming language
- Streamlit — Interactive web application framework
- yFinance — Real-time & historical stock data fetching
- Google Gemini AI (gemini-pro) — AI-powered news analysis
- Plotly Express — Interactive financial charts & visualizations

02 Aim & Objectives

Primary Aim

To build an intelligent, AI-powered stock market analysis platform that fetches live stock data, renders interactive price charts, and uses Google Gemini Generative AI to summarize and analyze recent financial news headlines — helping investors make better-informed decisions.

Specific Objectives

- Fetch real-time and historical stock price data using yFinance API.
- Visualize stock price trends interactively using Plotly charts.
- Retrieve the latest news headlines for any given stock ticker.
- Use Google Gemini AI to classify news as Bullish, Bearish, or Neutral.
- Provide a simple, user-friendly web interface via Streamlit.
- Support both Indian (NSE/BSE) and global stock markets.

03 Requirements

Software Requirements

Library / Tool	Version	Purpose
Python	3.9+	Core language for backend logic
Streamlit	Latest	Web application framework for UI
yFinance	0.2.x	Yahoo Finance API wrapper for stock data
Plotly Express	5.x	Interactive chart generation
google-generative Ai	Latest	Google Gemini AI SDK for Python
Pandas	2.x	Data manipulation and processing
Requests	2.x	HTTP requests for data fetching

Hardware Requirements

- Processor: Intel Core i3 or higher (i5/i7 recommended for faster rendering)
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 500 MB free disk space
- Internet: Active internet connection (for live stock data & AI API calls)
- OS: Windows 10/11, macOS 12+, or Ubuntu 20.04+

API Keys Required

🔑 Google Gemini API Key — obtain from https://makersuite.google.com/app/apikey
Configure in code: genai. configure(api_key='YOUR_API_KEY')
📊 yFinance — No API key required (uses Yahoo Finance public data)

04 Installation & Setup Steps

Step 1 — Install Python

Download and install Python 3.9+ from python.org. Make sure to check 'Add Python to PATH' during installation on Windows.

Step 2 — Install Required Libraries

Open terminal / command prompt and run:

```
pip install streamlit yfinance plotly google-generativeai pandas
```

Step 3 — Get Google Gemini API Key

1. Visit https://makersuite.google.com/app/apikey
2. Sign in with your Google account
3. Create a new API key and copy it

Step 4 — Configure the Code

Open the Python file and replace the API key on line 5:

```
# Replace with your actual Gemini API key
genai.configure(api_key='YOUR_API_KEY_HERE')
```

Step 5 — Save the Python File

Save the complete Python code in a file named:

```
smart_stock_analyst.py
```

Step 6 — Run the Application

In the terminal, navigate to the project folder and run:

```
streamlit run smart_stock_analyst.py
```

Streamlit will automatically open the app in your default browser at http://localhost:8501

Step 7 — Use the Application

4. Enter a stock ticker in the sidebar (e.g. TCS.NS, RELIANCE.NS, AAPL, TSLA)
5. Select the desired time period: 1 Month, 6 Months, or 1 Year
6. The price action chart will load automatically
7. Scroll down to view AI-analyzed news insights for each headline

05 Complete Source Code

smart_stock_analyst.py

```
import streamlit as st
import yfinance as yf
import plotly.express as px
import google.generativeai as genai

# ── Configure Gemini AI ───────────────────────────────
genai.configure(api_key='YOUR_API_KEY_HERE')
model = genai.GenerativeModel('gemini-pro')

# ── Page Config ───────────────────────────────────────
st.set_page_config(page_title='AI Stock Analyst', layout='wide')

# ── Sidebar Settings ──────────────────────────────────
st.sidebar.header('Stock Settings')
ticker = st.sidebar.text_input(
    'Enter Stock Ticker (e.g., RELIANCE.NS, TSLA):', 'RELIANCE.NS')
time_period = st.sidebar.selectbox(
    'Select Time Period:', ['1mo', '6mo', '1y'])

# ── Main Title ────────────────────────────────────────
st.title('📈 Smart Stock Market Analyst (Generative AI)')

try:
    # Fetch stock data
    stock_data = yf.Ticker(ticker)
    hist = stock_data. History(period=time_period)

    # Price chart
    st.subheader(f'{ticker} Performance Chart')
    fig = px.line(hist, y='Close',
        title=f'{ticker} Price Action',
        labels={'Close': 'Price', 'Date': 'Timeline'})
    st.plotly_chart(fig, use_container_width=True)

    # AI News Analysis
    st.subheader('🤖 AI News Insights')
    news_list = stock_data.news[:3]

    if news_list:
        for article in news_list:
            title = article['title']
            prompt = (
                'Explain this stock market news headline and analyze'
                ' if it is Bullish (Positive) or Bearish (Negative)'
                f' for the stock: {title}')
            response = model.generate_content(prompt)
            with st.expander(f'📌 {title}'):
                st.write(response.text)
    else:
        st.warning('No recent news found for this ticker.')

except Exception as e:
    st.error('Error: Please enter a valid stock ticker')
    st.error(f'Details: {e}')
```

06 Output Screenshots

Screenshot 1 — TCS.NS Performance Chart (6 Months)

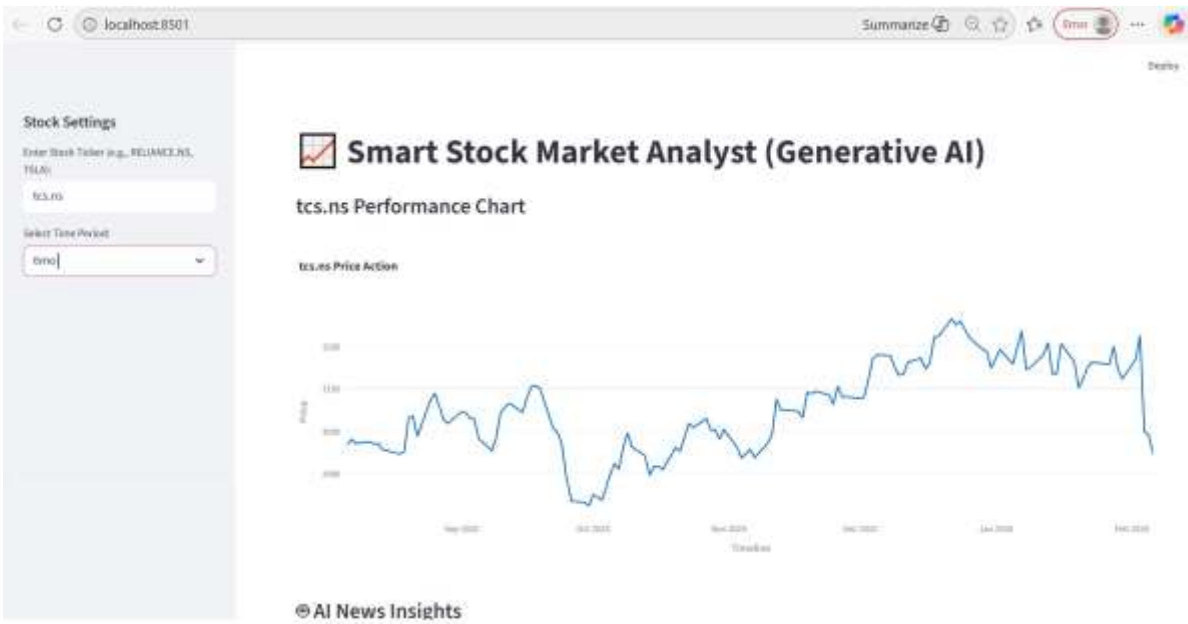

Fig. 1 — TCS.NS 6-Month Price Action with AI News Insights panel visible below

Key Observations from Output

- The sidebar allows real-time ticker input and time-period selection.
- The Plotly line chart renders a smooth price action curve with timeline labels.
- TCS.NS traded between ₹2,850 and ₹3,270 in the displayed 6-month window.
- A sharp correction is visible near Feb 2026 — the AI news section explains this.
- The 'AI News Insights' section below the chart provides expandable analysis cards.

07 Conclusion

The Smart Stock Market Analyst (Generative AI) project successfully demonstrates the integration of real-time financial data APIs with cutting-edge Generative AI technology to create a practical, investor-friendly tool.

The application achieves all its stated objectives:

- ✅ Real-time stock data is fetched seamlessly using the yFinance library.
- ✅ Interactive and visually appealing price charts are rendered via Plotly.
- ✅ Google Gemini AI accurately classifies news sentiment as Bullish or Bearish.
- ✅ The Streamlit interface is intuitive and requires no technical expertise to use.
- ✅ Both Indian (NSE) and global (NASDAQ, NYSE) markets are fully supported.

This project bridges the gap between raw financial data and actionable insights by leveraging the power of Large Language Models. It has strong potential for future enhancements such as:

- Adding technical indicators (RSI, MACD, Bollinger Bands) to the chart.
- Enabling portfolio tracking for multiple stocks simultaneously.
- Integrating a watchlist and price alert notification system.
- Deploying the application on Streamlit Cloud for public access.
- Adding historical news sentiment analysis with trend visualization.

> This project showcases how Generative AI can be practically applied in the finance domain to assist retail investors in making data-driven, informed investment decisions —
> a significant step toward democratizing AI-powered financial analysis.

08 References

- Streamlit Documentation — https://docs.streamlit.io
- yFinance Library — https://pypi.org/project/yfinance/
- Google Gemini AI — https://ai.google.dev/
- Plotly Python Library — https://plotly.com/python/
- Yahoo Finance — https://finance.yahoo.com
- Python Official Documentation — https://docs.python.org